jstal-tagger – user manual

jstal-tagger is a small web application that checks and maintains the ID3 tags of the MP3 files in your music folder, so the files are ready for the jukebox application. It runs in Docker on your own server (port 5080). There is no login: use it on your own network only, never on the internet.

1. What it does

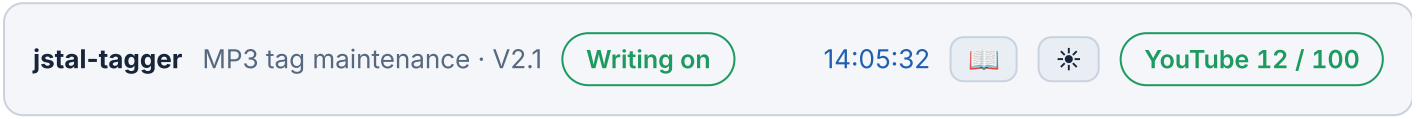
- **Browse** the music folder and **list** the MP3 files with their tags in a grid.
- **Edit** tags one cell at a time. Edits are kept as **drafts** in the browser (yellow cells) until you save them.
- **Look up** missing data online: year, cover, comment, YouTube link and lyrics.
- **Listen** to a song in the browser before you decide about its tags. V2.1
- **Save** the drafts of the selected songs, after a confirmation window that shows exactly what changes.
- **Wipe** chosen tag fields from the selected songs.
- **Report** the result per file, **export** the tags as JSON.

Every write first makes a verified backup copy and changes only the fields you approved; the audio data of a file never changes (see 15. Safety).

What it does **not** do: rename or move files, convert audio, write anything to files outside the music folder, or write anything at all while the app is in read-only mode.

2. Starting and the header

Open `http://<server>:5080` in a browser (Chrome, Edge or Firefox; after an update press `Ctrl` + `F5` once so the browser loads the new page files).



The header.

Item	Meaning
Writing on / Read-only	Whether the server may write to the files. **Read-only** means `ENABLE_TAG_WRITES=false` or a read-only music folder: you can still list, edit drafts, look up data and check plans, but nothing is written.
YouTube 12 / 100 V2.1	YouTube searches used today and the daily limit. Only shown when a YouTube key is configured. Click it to refresh. See 11. The YouTube counter.
Clock	The time in the browser.
📖 V2.1	Opens this manual in a new tab.
☀ / 🌙	Switches between the dark and the light theme (remembered by the browser).

3. Settings (.env)

All settings are in the `.env` file next to `docker-compose.yml` (an example is in `.env.example`). After a change run `docker compose up -d --force-recreate`.

Setting	Meaning
MUSIC_HOST_PATH	The music folder on the server (mounted as `/music`).
BACKUP_HOST_PATH	The backup folder, **outside** the music folder (mounted as `/backups`).
ENABLE_TAG_WRITES	`false` (default) = read-only. Writing needs `true` **and** `MUSIC_READ_ONLY=false`.
MUSIC_READ_ONLY	`true` (default) mounts the music folder read-only.
KEEP_BACKUPS	`false` (default): the backups of a save in which every file was saved and verified are removed afterwards. Wipe backups and saves with a failed file are always kept. `true` keeps every backup.
EDIT_MODE_DEFAULT V2.1	`true` (default): the page starts with **Edit tags** switched on. `false`: it starts switched off, as in V2.
LOOKUP_USER_AGENT	Required by MusicBrainz, LRCLIB and Megalobiz: `jstal-tagger/2.1 (mailto:you@example.com)` with your real contact.
YOUTUBE_API_KEY	YouTube Data API v3 key. Empty = the YouTube lookup is off (shown as "not configured").
YOUTUBE_REGION	Two-letter country code for the YouTube search, default `BE`.
YOUTUBE_PREFERRED_CHANNELS	Channel names that are ranked higher, separated by `|`.
YOUTUBE_DAILY_LIMIT V2.1	Searches per day before the app refuses further YouTube lookups, default `100` (the free quota).
DISCOGS_TOKEN	Discogs personal access token. Empty = the Discogs lookup is off.
DISCOGS_MASTER_OVERRIDES	JSON `{"artist|title": master id}` for songs that Discogs matches wrongly.
MEGALOBIZ_API_KEY V2.1	Optional. Megalobiz works without a key (30 requests a minute, 1000 a day); a free key raises the limits.

4. Folders and listing

The folder bar shows where you are (the breadcrumb 🎵 **Music › Oldies**) and the sub-folders as chips.

- Click a **folder chip** to open it; click a part of the breadcrumb or ↑ to go back.
- Tick **Include subfolders** to list the songs of all sub-folders as well.

- Click **List MP3 files**. Every file is read and hashed, so a large folder takes a moment (about 100 songs in half a second). Files that cannot be read get a red **Read error**.

Only files with the extension `.mp3` are listed. Symbolic links and files outside the music folder are ignored.

> Changing the folder or listing again discards unsaved drafts (the page asks first).

5. The song grid

Column	Shows
Filename	Click it for **all** ID3 tags of the file, the drafts and the SHA-256 (with a JSON download).
Listen V2.1	▶ plays the song (see 7).
Artist, Title, Album	TPE1, TIT2, TALB. Several values in one tag are shown as "Toto / Guest Singer".
Year	TDRC (ID3v2.4) or TYER (ID3v2.3). Four digits.
BPM, Track, Genre	TBPM, TRCK, TCON.
Duration	The **measured** audio time. Editing it only changes the stored TLEN value (milliseconds).
Cover	A thumbnail of the embedded front cover (APIC). Click it to see it large.
Lyrics	**LRC** (synchronised, lines like `[01:23.45] text`), **Plain**, or nothing (USLT).
YouTube URL (TVSHOW)	The link the jukebox uses for the video (custom tag TXXX:TVSHOW).
VIDEO	Tick box: ticked = TXXX:VIDEO is "Yes", unticked = the tag is removed.
Comment	COMM (English, without description).
Other tags	The number of ID3 frames that have no column (URLs, key, ReplayGain, custom TXXX, unknown frames). Click the number for the list.
Relative path, Read error	Where the file is, and why it could not be read.

- **Sort**: click a column title (again for the other direction).
- **Filter loaded songs**: only songs that contain *all* typed words (in any column) stay visible. The filter does not change the selection.
- **Columns** hides or shows columns; the choice is remembered by the browser.
- The counters under the toolbar show the number of files, how many are shown, selected, with drafts, and with read errors.

6. Selecting songs

Saving, wiping and exporting work on the **selected** songs (blue rows).

- Tick the box at the start of a row. The box in the column header selects or deselects all *shown* songs.
- **Select all** / **Deselect all** in the toolbar also include songs hidden by the filter.
- A song with a draft is selected automatically, so **Save tags** includes it.
- After a successful save the saved songs are deselected, so a wipe that follows cannot hit them by accident.

7. Listening to a song V2.1

Click ▶ in the **Listen** column. A player bar appears above the grid with the artist and title, the browser's own play / pause, position and volume controls. The same button (now ⏸) pauses and resumes; click ▶ on another row to switch songs; ✕ closes the player.

In the tag editor the ▶ **Listen** button next to the file name does the same, which is handy when you check lyrics or a YouTube result.

Listening is read-only: the server streams the file as it is, nothing is changed and the song keeps playing while you edit, save or wipe. The browser can seek (jump) inside the song.

8. Editing tags (drafts)

Edit mode

Editing is possible when the **Edit tags** switch is on. Since V2.1 the page starts with it on (`EDIT_MODE_DEFAULT`). The grid gets a yellow border and editable cells show a dashed outline when you hover.

Changing a cell

1. **Double-click** a cell (or click it and press `Enter`). The tag editor opens with the current value.
2. Type the new value, or use **Lookup data** (see 10).
3. **OK** makes a **draft**: the cell turns yellow. Nothing is written yet. **Cancel** or `Esc` closes without a draft.

- OK without a change makes no draft; typing the old value back removes the draft.
- **Clear field** makes a draft that **deletes** the field when saving ("(cleared)").
- When a cell has a draft, the editor also shows the value that is saved in the file.
- **VIDEO** is a tick box in the grid itself (in edit mode): ticked = Yes, unticked = the tag is removed.
- **Duration**: the editor changes only the stored TLEN value (milliseconds); the measured time stays.
- **Year**: four digits. **Track**: a number or number/total (6/12). **BPM**: a number.
- **Lyrics**: saving replaces all lyrics of the file, also synchronised (SYLT) ones.
- A tag with several values (shown with " / ") keeps them when you press OK without a change; a changed text is saved as one value.

> **Drafts live in the browser.** They are lost when you reload or close the page (the browser warns), when you change the folder or list again. **Discard drafts** throws them away on purpose.

9. Covers

Double-click the Cover cell (edit mode) or click the thumbnail. In the cover editor you can:

- **Choose a picture on this computer** (JPEG, PNG or WEBP, up to 25 MiB). Big pictures are made smaller in the browser (at most 600 × 600, never enlarged); the server stores a JPEG and replaces *all* embedded pictures with that one front cover.
- Use **Lookup data** to find a cover on MusicBrainz (Cover Art Archive) or Discogs, click **Use** on a result and OK. The cover is downloaded by the server (artwork sites only) as a 600 px JPEG draft.
- **MusicHoarders** opens in a new tab: download a cover there, then choose it with "Choose a picture".
- **Clear field** removes all pictures when saving.

10. Lookup data (online sources)

For the fields **Year, Cover, Comment, YouTube URL and Lyrics** the tag editor has a **Lookup data** section.

1. Adjust the **Artist** and **Title** used for the search if needed (this does not change the tags).
2. Tick the sources and click **Lookup Data**. Each source shows its results in its own column.
3. Click **Use** on a result: it fills the field (or shows the cover). You can still change the text before OK.
4. Click **OK** to make the draft.

Field	Sources	Notes
Year	MusicBrainz, Discogs	The earliest year found is filled in already; check the *View source* link.
Cover	MusicBrainz / Cover Art Archive, Discogs, MusicHoarders (browser tab)	Front covers of matching releases.
Comment	MusicBrainz + Wikipedia, Discogs artist profile	A short artist / recording text.
YouTube URL	YouTube, Discogs release videos	Public, embeddable music videos ranked on title, artist, duration and "official" wording. Live and concert uploads are skipped. Counted by the daily counter.
Lyrics	LRCLIB, Megalobiz V2.1	Synchronised (LRC) lyrics are preferred over plain text. Megalobiz only returns LRC sheets.

Megalobiz V2.1

Megalobiz (megalobiz.com) is a large collection of user-made synchronised lyrics. The app uses its public API: it searches on artist, title and duration and fetches the best matching sheets (at most 3). A

sheet is only shown when artist and title match closely (90 %) and the duration is within 3 seconds; sheets without timestamps are skipped. Because the sheets are timed by hand, play the song (▶ Listen) and check that the lines keep up before you save. The *View source* link opens the sheet's page on Megalobiz.

> Matches are heuristic for every source. Always look at the source link before you save; "Use" and OK only make a draft.

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11. The YouTube counter V2.1

The YouTube Data API gives 10 000 quota units a day; one search costs 100 units, so about **100 searches a day** are possible. The header shows **YouTube n / 100**: the searches used today and the limit.

- The badge is green, turns **orange** when fewer than about 15 % are left and **red** when the limit is reached.
- When the limit is reached the app does not call YouTube any more: the lookup column says why, a message pops up, and the YouTube source is unticked in the lookup section (it says "daily limit reached").
- If Google itself reports that the quota is used up (it also counts calls from other programs with the same key), the badge shows that too.
- Google resets the quota at **midnight Pacific time** (about 09:00 in Belgium); the badge's tooltip shows the local time. Click the badge to refresh it.
- A result that is still in the cache (six hours) costs nothing.
- The counter is saved in `.jstal-tagger/youtube-quota.json` in the backup folder, so it survives a restart of the container. The daily limit can be changed with `YOUTUBE_DAILY_LIMIT`.

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12. Saving

1. Make drafts (yellow cells). The **Save tags (n)** button shows how many selected songs have drafts.
2. Click it. The server checks every file and value and shows a **confirmation window**: per song a table *Field · Now in the file · New*; covers as pictures. The raw plan is under "Technical details".
3. Click **Save these changes**. A progress bar shows "Saving 3 / 8".
4. Saved songs are read again, their drafts disappear and they are deselected. A message shows the result; if a file failed, the **Last report** opens.

- At most 100 songs per save. A prepared plan is valid for 30 minutes and can be used once.
- A file that changed since it was listed is refused ("File changed since it was listed"): list the folder again.
- **Cancel — nothing is written** closes the window without writing.
- In read-only mode you can prepare and check a plan, but the save button stays disabled.

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13. Wiping tags

Wiping removes chosen fields from the selected songs. Save or discard your drafts first.

1. Select the songs and click **Wipe tags…**.

2. Tick the fields to remove. **Artist, Title and Album are unticked by default** (protected). **Select all** and **Deselect all** V2.1 tick or untick every field at once; **Default choice** restores the default. The line at the bottom shows how many fields are ticked and warns when Artist, Title or Album is among them.
3. Click **Prepare wipe…**. The confirmation window lists per song the exact ID3 frames that are removed and the fields that stay protected.
4. The confirmation text **WIPE SELECTED TAGS** is already filled in V2.1. Check the list, then click **Wipe these tags**. (If you change the text, the button is disabled until it matches again.)

- **Other tags** removes every frame that has no column: URLs, key, ReplayGain, custom TXXX and unknown frames. A file with APEv2 tags (not ID3) is refused for that, rather than leaving hidden tags behind.
- **Duration** only removes the stored TLEN value; **VIDEO** removes the TXXX:VIDEO tag.
- Wipe backups are always kept (see 15).

> A wipe is a real change to your files. The protection is the exact list in the confirmation window — read it.

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14. Report and export

- **Last report** shows the result per file of the last save or wipe since the container started: saved / failed, the error text, the batch name and where the backups are. **Download JSON** saves it. The full report of a kept batch is also in `BACKUP_HOST_PATH/<batch>/write-report.json`.
- **Export selected** downloads the tags of the selected songs as a JSON file (nothing is written to the songs).
- Clicking a **filename** shows all ID3 tags of that file; the JSON of one song can be downloaded there too.

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15. Safety, backups, restore

- Nothing is written unless `ENABLE_TAG_WRITES=true` **and** the music folder is writable.
- Every save or wipe is **prepared** first: the server checks every file and value and shows the exact plan.
- Per file: a verified backup copy is made first; the change is made on a temporary copy; it is checked that only the approved frames changed and that the audio part is byte-for-byte the same; only then the original is replaced (same permissions and owner). A failure leaves the original untouched and keeps the failed attempt next to the backup.
- A failure is per file: the other files of the batch are still saved.
- Backups are in `BACKUP_HOST_PATH/<batch>/<path>`; batches are named `save-…` or `wipe-…`. With `KEEP_BACKUPS=false` the backups of a fully successful *save* are removed afterwards; wipe backups and batches with a failed file are always kept.
- Only ID3v2.3 / ID3v2.4 (and untagged) files are written. ID3v1 copies of changed fields are updated.
- Refused: symbolic links, hard-linked files, files larger than 512 MiB, paths outside the music folder.
- One save or wipe at a time. Do not move or edit the files with another program while a save runs, and do not stop the container during a save.

Restore (manual)

1. Stop the container: `docker compose stop`.
2. Find the batch in the backup folder (the **Last report** or `write-report.json` names it) and copy the backup file back over the song.

3. Start the container and list the folder again. There is no restore button.

16. Messages and troubleshooting

Message	What to do
Read-only (header)	Set `ENABLE_TAG_WRITES=true` and `MUSIC_READ_ONLY=false` in .env, then `docker compose up -d --force-recreate`.
File changed since it was listed	Another program changed the file. List the folder again and make the draft again.
Same-origin request required	Open the app directly with its own address and port (not through another page or proxy that changes the address).
Set LOOKUP_USER_AGENT ...	MusicBrainz, LRCLIB and Megalobiz need it. Fill it in .env with your real contact.
Not configured (Discogs / YouTube)	Add `DISCOGS_TOKEN` / `YOUTUBE_API_KEY` to .env.
The daily limit of 100 YouTube searches is reached	Wait for the reset (about 09:00 Belgian time). Cached results still work.
Megalobiz rate limit reached	Wait a minute (30 requests a minute without a key) or add a free `MEGALOBIZ_API_KEY`.
APEv2 tags found	The file has APE tags next to ID3. Remove them with another tool first, or untick "Other tags" in the wipe.
The request is too large	Save fewer covers at once.
This file cannot be played in the browser	The file is damaged or not a real MP3. Click the filename: the Read error says more.
A save or wipe is still running	Wait for the progress bar to finish. After a page reload the app re-attaches to the running job.
Lost the connection to the save job	Reload the page; check the Last report. Files are never half-written.

Useful checks on the server (nothing is written):

```
docker compose logs -f jstal-tagger
docker compose run --rm --no-deps --entrypoint python jstal-tagger -c "import app;
print('Application imports successfully.')"
docker compose run --rm --no-deps --entrypoint python jstal-tagger tags.py --self-test
"/music/SOME SONG.mp3"
```

17. Quick reference

Mouse

Folder chip / breadcrumb	open a folder
Column title	sort
Filename	all tags of the file
▶	listen / pause
Double-click a cell	edit it (edit mode)
Cover thumbnail	show it large / edit it
"Other tags" number	list of the other frames
YouTube badge	refresh the counter

Keyboard

`Enter` **on a cell**	edit it (edit mode)
`Enter` **in the editor**	OK (single-line fields)
`Esc`	close a window without changes
`Ctrl` + `F5`	reload after an update

Files of the app

app.py	web server
tags.py	reading / safe writing of tags
lookups.py	online sources
templates/, static/	the page, this manual, script, style